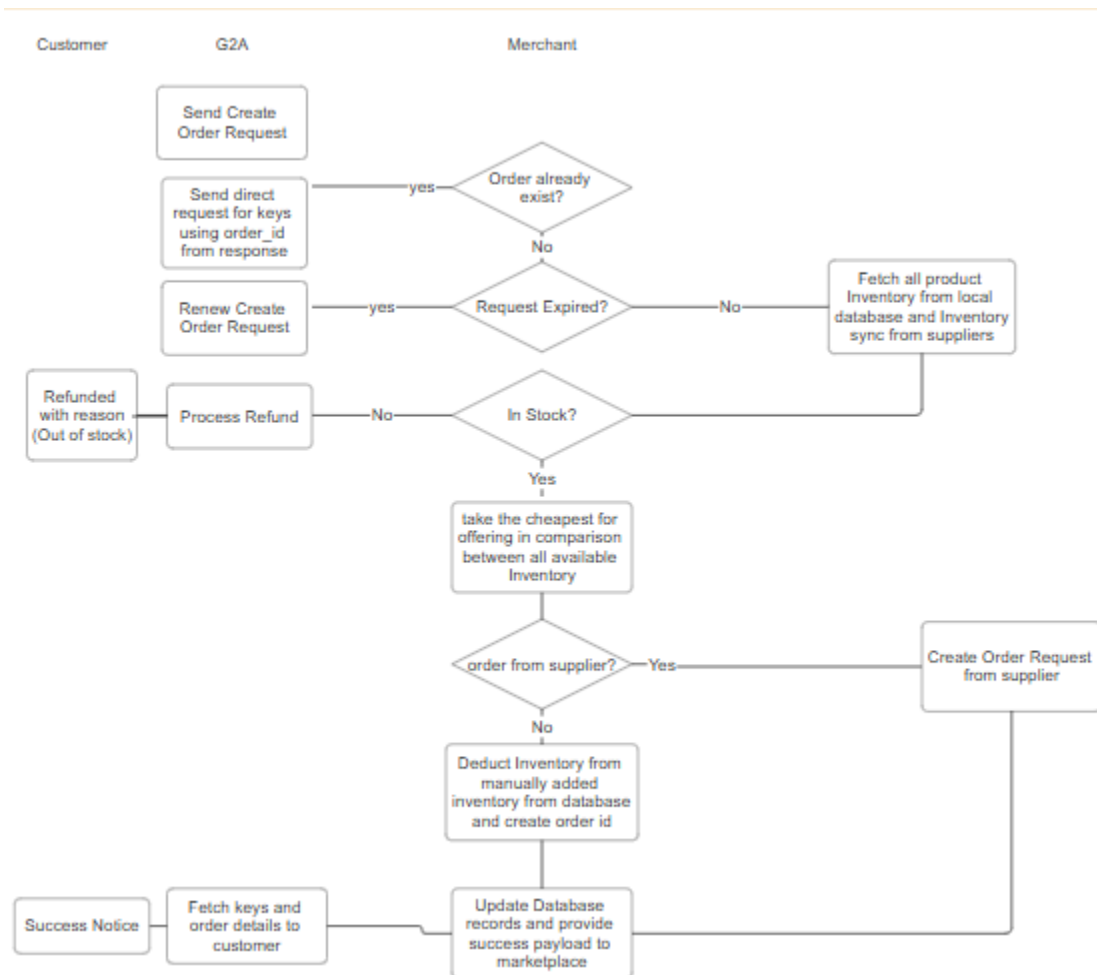


Card Cove (Technical design document)

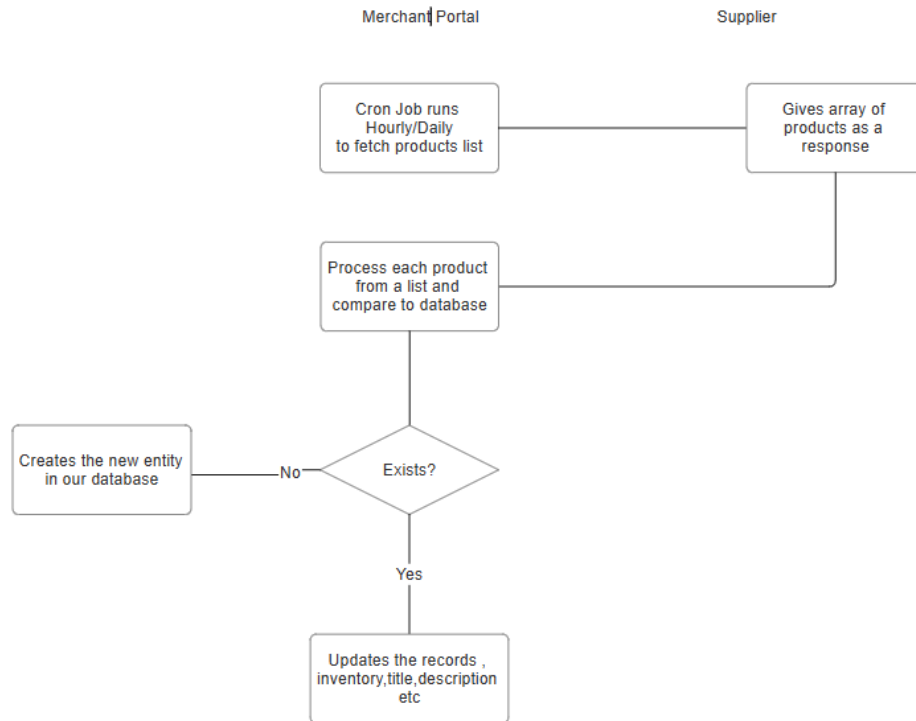
1. Application Architecture Overview

- **Marketplaces** (starting with G2A)
- **Suppliers** (starting with Vannanam and Joker)
- **Internal DB** for inventory & pricing (suppliers via API and Manually added products)

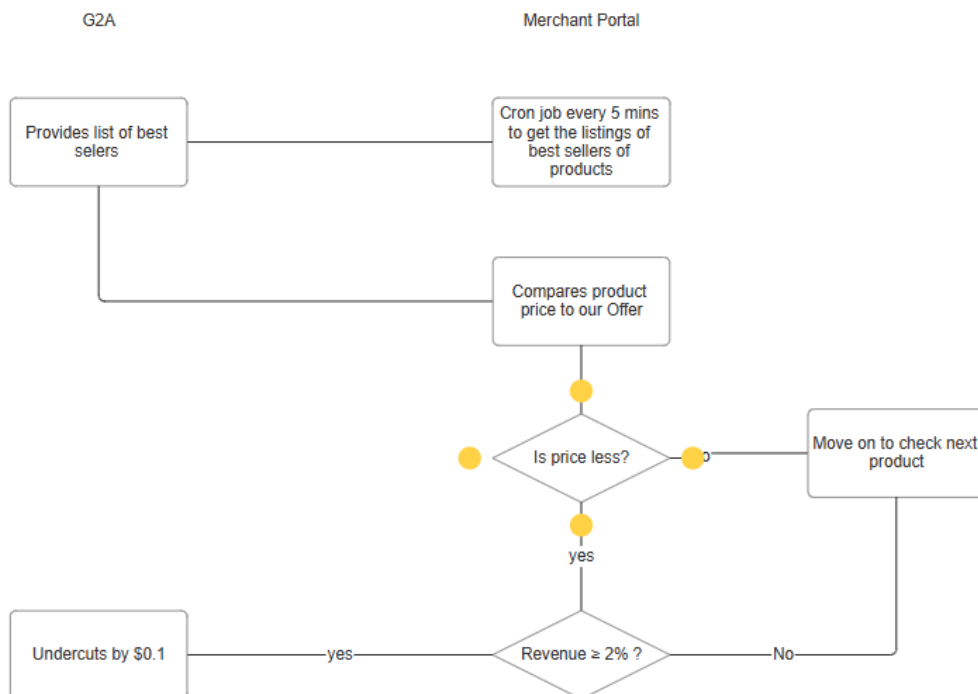
Note: Suppliers and Marketplace Sandbox required. These Observations are from online documentation , Which may vary from actual responses.



1) Order placing Workflow



2) Inventory Sync from Supplier



3) Price Anomaly Workflow

2. Authentication Flow

2.1 Sign-In Page

- **Fields:** Username, Password
- **Validation:**
 - Minimum 12 characters
 - Uppercase, lowercase, numbers
- **Security:**
 - Lock after 5 failed attempts
 - 15-minute session timeout
 - **2FA via Google Authenticator**

3. Dashboard Page

Overview Reports:

Sales Report	SKU, Sales Qty, Revenue, Margin
Supplier performance	Supplier name, On-time rate, Cost fluctuations
Inventory Snapshot	Daily/Live SKU, Quantity, Cost,
Pricing anomaly report	Price difference between suppliers

Note: Other reports will be integrated on request.

4. Local Inventory Management

4.1 Products Page

- **Actions:**
 - Search Products by SKU, name, category
 - Add/Amend/Delete product manually
 - Export product list

5. Order Management Page

- **Actions:**
 - Search/filter orders
 - View order status

6. Marketplace Integration – G2A

G2A APIs Used:

- POST /create-offer
- GET /get-products
- GET /get-prices
- GET /get-best-sellers
- GET /get-offer-list

- POST /update-offer-visibility
- POST /set-offer-price

8. Smart Pricing Engine

Features:

- Auto-checks competitor prices every 5 minutes
- Undercuts by \$0.1 if profit $\geq$ 2%
- Syncs adjusted price to:
 - G2A Offers
 - Local DB

Note (Question) : For example we have 3 different prices from 3 best sellers for one product are we going to reduce our price if even 1 seller is selling less than us or we are going to take average price of all sellers to make our decision?

Inventory and Order Placement Workflow:

FrontEnd Pages

Feature	Description
Product Catalog	View all synced products & pricing
Offer Creation	UI to publish offer to G2A
Order Dashboard	Monitor incoming orders and fulfillment

PHASE 1: Supplier Inventory Sync

Goal: Sync external supplier inventory (Vannanam) to your DB.

Steps:

Fetch Product List from Supplier

- **API:** GET <https://developers.thereward.store/products>
- Response payloads:

```
{  
  "id": 54,  
  "name": "TrialTaste Sampler- URL",  
  "details": "<p>This is a sample Voucher for The Reward Store.</p>",  
  "how_to_use": "This is a sample Voucher for The Reward Store.",  
  "image_url":  
"https://imagedelivery.net/7zckf4ay8vHyPu-s4kSwMA/bfc3fa3e-91a7-49c1-ce5b-8dcf8  
bf3b600/public",  
  "category": "Gift Card",  
  "sub_category": "E-Commerce",  
  "delivery_time": 3,  
  "validity": 12,  
  "delivery_type": "REALTIME",  
  "terms_and_conditions": "This is a sample Voucher for The Reward Store.",  
  "redemption_type": "ONLINE",
```

```
"denominations": [  
  {  
    "discount": -4,  
    "min": 100,  
    "max": 100  
  }  
,  
  "country": "SGP",  
  "currency": "SGD"  
},
```

- **Schedule:** c CRON job /manual on demand

Save or Update Local Database

- Compare incoming product SKU with local entries
- Update:
 - price
 - stock
 - title/description
- Insert new entries if missing

PHASE 2: Offer Creation to G2A Marketplace

Goal: Select synced products and create marketplace offers.

Operator Actions:

- Use Admin UI to:
 - Create Offers

Steps:

1. Create G2A Offer

- **API:** POST /create-offer
- Payload includes:
 - product_id
 - title, description
 - price
 - stock (based on available supplier inventory or manual)

2. Track G2A Offer

- Store G2A offer_id in your DB
- Track link to internal product & supplier product

Result: Product is now listed on G2A for sale.

PHASE 3: G2A Price & Inventory Automation

Goal: Continuously keep your prices competitive.

Steps (Cron every 5 mins):

1. Check Market Prices

- **API:** GET /get-prices per product

2. Apply Pricing Logic

- Compare your price with lowest
- If margin > 2% → Set your price = competitor - 0.1
- Else → skip update

3. Update Offer Price

- **API:** POST /set-offer-price

4. Check Key Inventory / Visibility

- If no keys locally or at supplier → **hide offer**
- **API:** POST /deactivate-offer

Result: Your offers are always competitive and visible only if you have stock.

PHASE 4: G2A Order Placement

1. Receive Order from G2A

- **Method:** Set up a webhook or periodically poll G2A's order endpoint to detect new orders.
- **Endpoint:** GET /orders
- **Action:**
 - Retrieve order details, including:
 - Order ID
 - Product SKU
 - Quantity
 - Customer information

2. Check Internal Inventory

- **Action:**
 - Query your internal database to determine if the ordered product is available in your local inventory.
 - If available:
 - Reserve the product.
 - Proceed to deliver the product to the customer.
 - If not available:
 - Proceed to the next step to source the product from suppliers.

3. Compare Supplier Prices

- **Suppliers:** TheReward.Store (Vannanam), Joker, etc.
- **Endpoint:** For TheReward.Store, use GET /products
- **Action:**
 - Fetch the latest product listings and prices from each supplier.
 - Identify the supplier offering the best price for the required product.

4. Place Order with Selected Supplier

- **Endpoint:** For TheReward.Store, use POST /orders
- **Payload:**
 - Product SKU
 - Quantity
 - Delivery details

- **Action:**
 - Send a purchase order to the selected supplier.
 - Receive confirmation and tracking details.

5. Update Internal Records

- **Action:**
 - Record the transaction details in your system, including:
 - Supplier name
 - Purchase price
 - Order ID
 - Delivery status

Tech Overview

FrontEnd: Next js (Framework of React js , Mature and modernized integration libraries available and SEO friendly).

Middleware: Express js (Lightweight API server)

Backend: Node js (Same language (JavaScript) across the stack, Handles asynchronous API calls (like G2A/Supplier APIs) very efficiently, and lightweight Setup.

Database: MySQL (Stronger data validation, relationships, and querying)

Card Cove Commercial website : Wordpress (5 page standard website)

Purchase of Hosting and Domain is responsibility of Card Cove , Development team will be responsible for deployment on any hosting environment.

Exception Handling:

Using Flat files for documenting each exceptions and Audit logs. Crucial Alerts can be generated via Card Cove SMTP.

